

CEE6501 Final Course Project: DSM-Based Structural Analysis Program

CEE 6501 Matrix Structural Analysis

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Project Overview

- Developed a general-purpose structural analysis program based on the Direct Stiffness Method (DSM)
- Analysis of 3D truss and 3D frame elements
- Support for multiple loading and boundary conditions
- Objective: Provide a flexible and reusable computational framework for structural analysis

Code Architecture

- Structured into three main components:

- **Input**

- **DSM implementation**

- **Output**



Input Format

- Structural model defined via **JSON input file**, Includes:
 - ✓ Node coordinates
 - ✓ Element connectivity and types
 - ✓ Material and section properties
 - ✓ Loads and boundary conditions

```
"materials": {  
  "steel": {  
    "E": 20.0  
  }  
},  
  
"sections": {  
  "A1": {  
    "A": 5000.0  
  }  
},
```

```
  "elements": {  
    "1": [1, 3, "A1"],  
    "2": [2, 4, "A1"],  
    .....  
  },  
  
  "nodal loads": {  
    "1": [0, -80, 0, 0, 0, 0],  
    "2": [0, -80, 0, 0, 0, 0],  
    "3": [0, -80, 0, 0, 0, 0],  
    .....  
    "75": [0, -80, 0, 0, 0, 0],  
    "76": [0, -80, 0, 0, 0, 0]  
  },  
  
  "member loads": {  
    "133": [{"type": "udl local", "qx": 0.0, "qy": -20.0, "qz": 0.0}],  
    "135": [{"type": "udl local", "qx": 0.0, "qy": -20.0, "qz": 0.0}],
```

Key Implementation Features

- Mixed element capability: 3D truss + 3D frame in a unified framework
- Additional functionalities:
 - ✓ Member end releases
 - ✓ Member loading
 - ✓ Support settlements
 - ✓ Temperature effects (uniform)
 - ✓ Fabrication error (length variation)

Validation of the Program

- Conducted validation using:

- ✓ One truss example
- ✓ Four frame example

- Method:

Comparison between analytical (hand) solutions and numerical results

- Outcome:

Good agreement confirms correctness of implementation

```
{ } validation_frame_baseline.json    U
{ } validation_frame_fabrication_err... U
{ } validation_frame_release.json    U
{ } validation_frame_support_settle... U
{ } validation_truss.json             M
```

Validation of the Program

■ Truss: A5_code Q1

=== ELEMENT RESULTS ===

E1 (Truss 3D)

u_global [mm,rad]: [0.000, -4.414, 0.000, 0.000000, 0.000000, 0.000000, 0.000, 0.000, 0.000, 0.000000, 0.000000, 0.000000]
v_local [mm,rad]: [-0.000, 0.000, -4.414, 0.000000, 0.000000, 0.000000, -0.000, 0.000, 0.000, 0.000000, 0.000000, 0.000000]
q_local [kN,kN·m]: [0.000, 0.000, 0.000, 0.000, 0.000, 0.000, -0.000, 0.000, 0.000, 0.000, 0.000, 0.000]

E2 (Truss 3D)

u_global [mm,rad]: [0.000, -4.414, 0.000, 0.000000, 0.000000, 0.000000, 2.914, -1.500, 0.000, 0.000000, 0.000000, 0.000000]
v_local [mm,rad]: [-3.121, 0.000, -3.121, 0.000000, 0.000000, 0.000000, -3.121, 0.000, 1.000, 0.000000, 0.000000, 0.000000]
q_local [kN,kN·m]: [0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000]

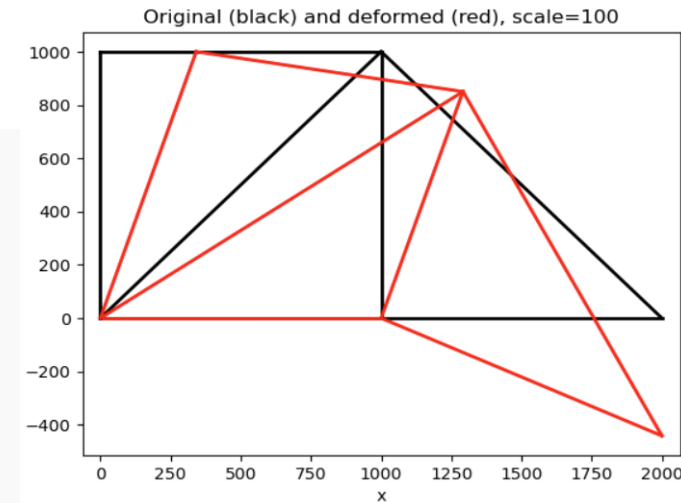
E3 (Truss 3D)

u_global [mm,rad]: [0.000, 0.000, 0.000, 0.000000, 0.000000, 0.000000, 2.914, -1.500, 0.000, 0.000000, 0.000000, 0.000000]
v_local [mm,rad]: [0.000, 0.000, 0.000, 0.000000, 0.000000, 0.000000, -1.500, 0.000, 2.914, 0.000000, 0.000000, 0.000000]
q_local [kN,kN·m]: [150.000, 0.000, 0.000, 0.000, 0.000, 0.000, -150.000, 0.000, 0.000, 0.000, 0.000, 0.000]

E4 (Truss 3D)

u_global [mm,rad]: [0.000, 0.000, 0.000, 0.000000, 0.000000, 0.000000, 0.000, 0.000, 0.000, 0.000000, 0.000000, 0.000000]
v_local [mm,rad]: [-0.000, 0.000, 0.000, 0.000000, 0.000000, 0.000000, 0.000, 0.000, 0.000, 0.000000, 0.000000, 0.000000]
q_local [kN,kN·m]: [-0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000]

E5 (Truss 3D)



	ele	i	j	L (mm)	N (kN)	sigma (MPa)	u_1 (mm)	u_2 (mm)	u_3 (mm)	u_4 (mm)	u_1' (mm)	u_2' (mm)	u_3' (mm)	u_4' (mm)	f_1' (kN)	f_2' (kN)	f_3' (kN)	f_4' (kN)
0	1	1	2	1000.0	0.0	0.0	0.0	-4.4	0.0	0.0	-0.0	4.4	0.0	0.0	-0.0	0.0	0.0	0.0
1	2	1	3	1414.2	0.0	0.0	0.0	-4.4	2.9	-1.5	-3.1	3.1	-3.1	-1.0	-0.0	0.0	0.0	0.0
2	3	2	3	1000.0	-150.0	-30.0	0.0	0.0	2.9	-1.5	0.0	0.0	-1.5	-2.9	150.0	0.0	-150.0	0.0
3	4	2	4	1000.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	5	3	4	1414.2	70.7	14.1	2.9	-1.5	0.0	0.0	-1.0	3.1	0.0	0.0	-70.7	0.0	70.7	0.0
5	6	3	5	1000.0	-50.0	-10.0	2.9	-1.5	3.4	0.0	-2.9	1.5	-3.4	0.0	50.0	0.0	-50.0	0.0
6	7	4	5	1000.0	0.0	0.0	0.0	0.0	3.4	0.0	0.0	0.0	0.0	-3.4	0.0	0.0	0.0	0.0

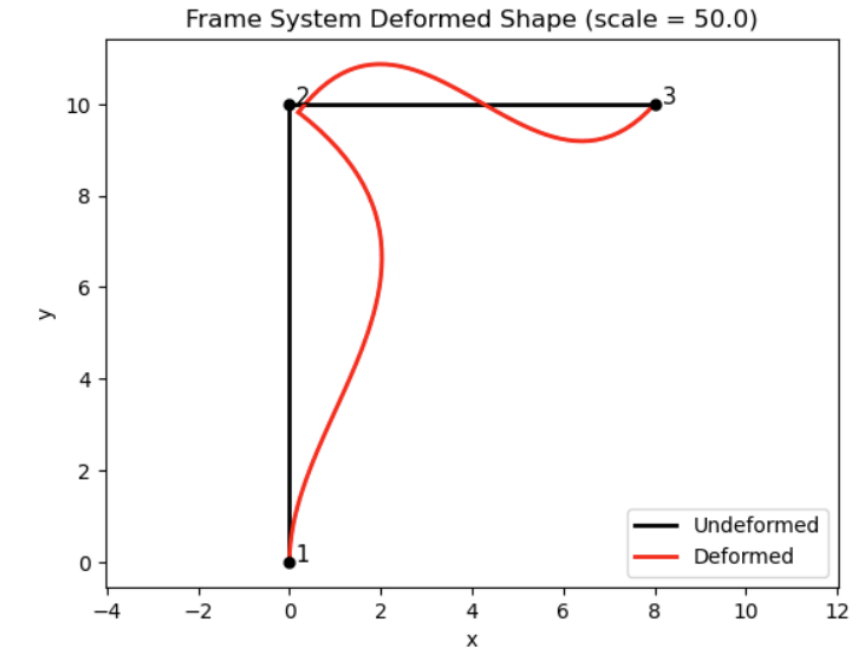
Validation of the Program

■ Frame_Baseline: A7_written Q2

Local element end forces and displacements:

① element 1: $u_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 9.55 \times 10^{-4} \\ -5.99 \times 10^{-4} \\ 2.54 \times 10^{-2} \end{bmatrix}$, Local displacement $u'_1 = T u_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -5.99 \times 10^{-4} \\ -9.55 \times 10^{-4} \\ 2.54 \times 10^{-2} \end{bmatrix}$ m, rad.

Local forces: $f'_1 = k_1 u'_1 + Q_{f_1} = \begin{bmatrix} 56.79 \\ 126.82 \\ 222.81 \\ -56.79 \\ 113.17 \\ -154.72 \end{bmatrix}$ kN, kN·m.



E1 (Frame 3D)

u_global [mm,rad]: [0.000, 0.000, 0.000, 0.000000, 0.000000, 0.000000, 0.955, -0.599, 0.000, 0.000000, 0.000000, 0.025391]
y_local [mm,rad]: [0.000, 0.000, 0.000, 0.000000, 0.000000, 0.000000, -0.599, -0.955, 0.000, 0.000000, 0.000000, 0.025391]
q_local [kN,kN·m]: [56.831, 126.815, 0.000, 0.000, 0.000, 222.802, -56.831, 113.185, 0.000, 0.000, 0.000, -154.651]

E2 (Frame 3D)

u_global [mm,rad]: [0.955, -0.599, 0.000, 0.000000, 0.000000, 0.025391, 0.000, 0.000, 0.000, 0.000000, 0.000000, 0.021201]
y_local [mm,rad]: [0.955, -0.599, 0.000, 0.000000, 0.000000, 0.025391, 0.000, 0.000, 0.000, 0.000000, 0.000000, 0.021201]
q_local [kN,kN·m]: [113.185, 56.831, 0.000, 0.000, 0.000, 154.651, -113.185, 18.169, 0.000, 0.000, 0.000, 0.000]

Validation of the Program

▪ Frame_support settlement: A10_written Q1

=== ELEMENT RESULTS ===

E1 (Frame 3D)

u_global [mm,rad]: [0.000, 0.000, 0.000, 0.000000, 0.000000, 0.010000, 5.938, -0.594, 0.000, 0.000000, 0.000000, 0.021819]
v_local [mm,rad]: [0.000, 0.000, 0.000, 0.000000, 0.000000, 0.010000, -0.594, -5.938, 0.000, 0.000000, 0.000000, 0.021819]
q_local [kN,kN.m]: [56.348, 128.793, 0.000, 0.000, 0.000, 238.718, -56.348, 111.207, 0.000, 0.000, 0.000, -150.787]

E2 (Frame 3D)

u_global [mm,rad]: [5.938, -0.594, 0.000, 0.000000, 0.000000, 0.021819, 5.000, -10.000, 0.000, 0.000000, 0.000000, 0.021111]
v_local [mm,rad]: [5.938, -0.594, 0.000, 0.000000, 0.000000, 0.021819, 5.000, -10.000, 0.000, 0.000000, 0.000000, 0.021111]
q_local [kN,kN.m]: [111.207, 56.348, 0.000, 0.000, 0.000, 150.787, -111.207, 18.652, 0.000, 0.000, 0.000, 0.000]

local forces:

element 1: $f_1' = k_1 u_1' + Q_1 \bar{F} = \begin{bmatrix} -56.88 \\ 8.79 \\ 38.7 \\ -56.88 \\ -8.79 \\ 49.18 \end{bmatrix} + \begin{bmatrix} 0 \\ 200 \\ 0 \\ 120 \\ -200 \end{bmatrix} = \begin{bmatrix} 56.88 \\ 128.79 \\ 238.7 \\ -56.88 \\ 111.21 \\ -150.82 \end{bmatrix}$

element 2: $f_2' = k_2 u_2' + Q_2 \bar{F} = \begin{bmatrix} 111.39 \\ 18.84 \\ 75.73 \\ -111.39 \\ -18.84 \\ 74.95 \end{bmatrix} + \begin{bmatrix} 0 \\ 37.5 \\ 75 \\ 0 \\ 37.5 \\ -75 \end{bmatrix} = \begin{bmatrix} 111.39 \\ 56.34 \\ 150.73 \\ -111.39 \\ 18.66 \\ -0.05 \end{bmatrix}$

Validation of the Program

▪ Frame_fabrication_error: A10_written Q3

E1 (Frame 3D)

u_global [mm,rad]: [0.000, 0.000, 0.000, 0.000000, 0.000000, 0.000000, 3.954, -3.599, 0.000, 0.000000, 0.000000, 0.025340]
v_local [mm,rad]: [0.000, 0.000, 0.000, 0.000000, 0.000000, 0.000000, -3.599, -3.954, 0.000, 0.000000, 0.000000, 0.025340]
q_local [kN,kN.m]: [56.743, 126.961, 0.000, 0.000, 0.000, 223.555, -56.743, 113.039, 0.000, 0.000, 0.000, -153.942]

E2 (Frame 3D)

u_global [mm,rad]: [3.954, -3.599, 0.000, 0.000000, 0.000000, 0.025340, 0.000, 0.000, 0.000, 0.000000, 0.000000, 0.021788]
v_local [mm,rad]: [3.954, -3.599, 0.000, 0.000000, 0.000000, 0.025340, 0.000, 0.000, 0.000, 0.000000, 0.000000, 0.021788]
q_local [kN,kN.m]: [113.039, 56.743, 0.000, 0.000, 0.000, 153.942, -113.039, 18.257, 0.000, 0.000, 0.000, 0.000]

element 1:

$$u_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 3.95 \times 10^{-3} \\ -3.6 \times 10^{-3} \\ 0.025 \end{bmatrix}$$

$$u_1' = T_1 u_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -3.6 \times 10^{-3} \\ 3.95 \times 10^{-3} \\ 0.025 \end{bmatrix}$$

element 2:

$$u_2 = u_2'$$

$$= \begin{bmatrix} 3.95 \times 10^{-3} \\ -3.6 \times 10^{-3} \\ 0.025 \\ 0 \\ 0 \\ 0.0217 \end{bmatrix}$$

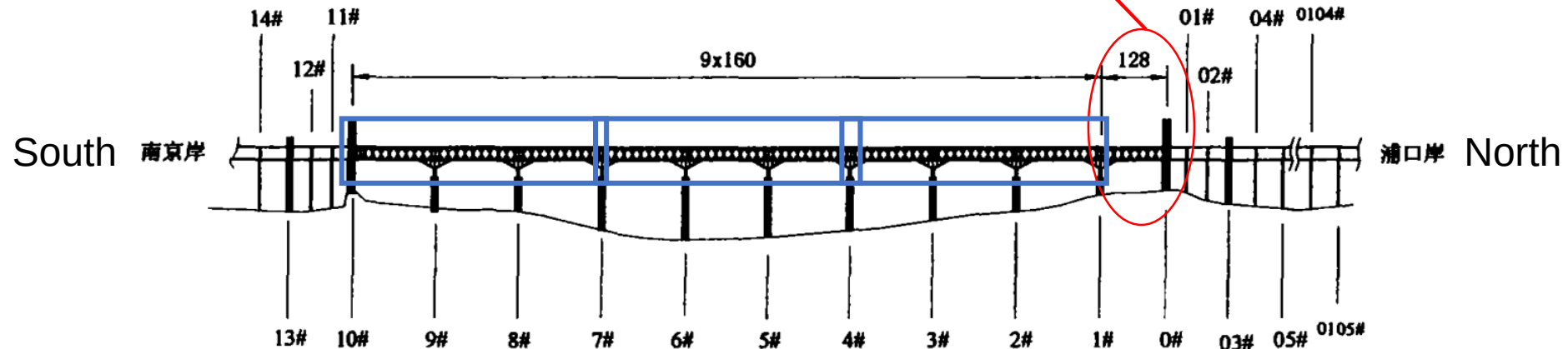
Validation of the Program

- Frame_member_release: A8_code

```
=== GLOBAL RESULTS ===
DOF      Type      Status      Disp (mm / rad)      Load (kN / kN.m)
  1      u_x      Fixed          0.000          -120.054
  2      u_y      Fixed          0.000           51.548
  3      u_z      Fixed          0.000           0.000
  4  theta_x      Fixed          0.000000         0.000
  5  theta_y      Fixed          0.000000         0.000
  6  theta_z      Fixed          0.000000        200.270
  7      u_x      Free           1.012         120.000
  8      u_y      Free          -0.544        -37.500
  9      u_z      Fixed          0.000           0.000
 10  theta_x      Fixed          0.000000         0.000
 11  theta_y      Fixed          0.000000         0.000
 12  theta_z      Free           0.000000         0.000
 13      u_x      Fixed          0.000        -119.946
 14      u_y      Fixed          0.000          23.452
 15      u_z      Fixed          0.000           0.000
 16  theta_x      Fixed          0.000000         0.000
 17  theta_y      Fixed          0.000000         0.000
 18  theta_z      Free          0.033886         75.000
```

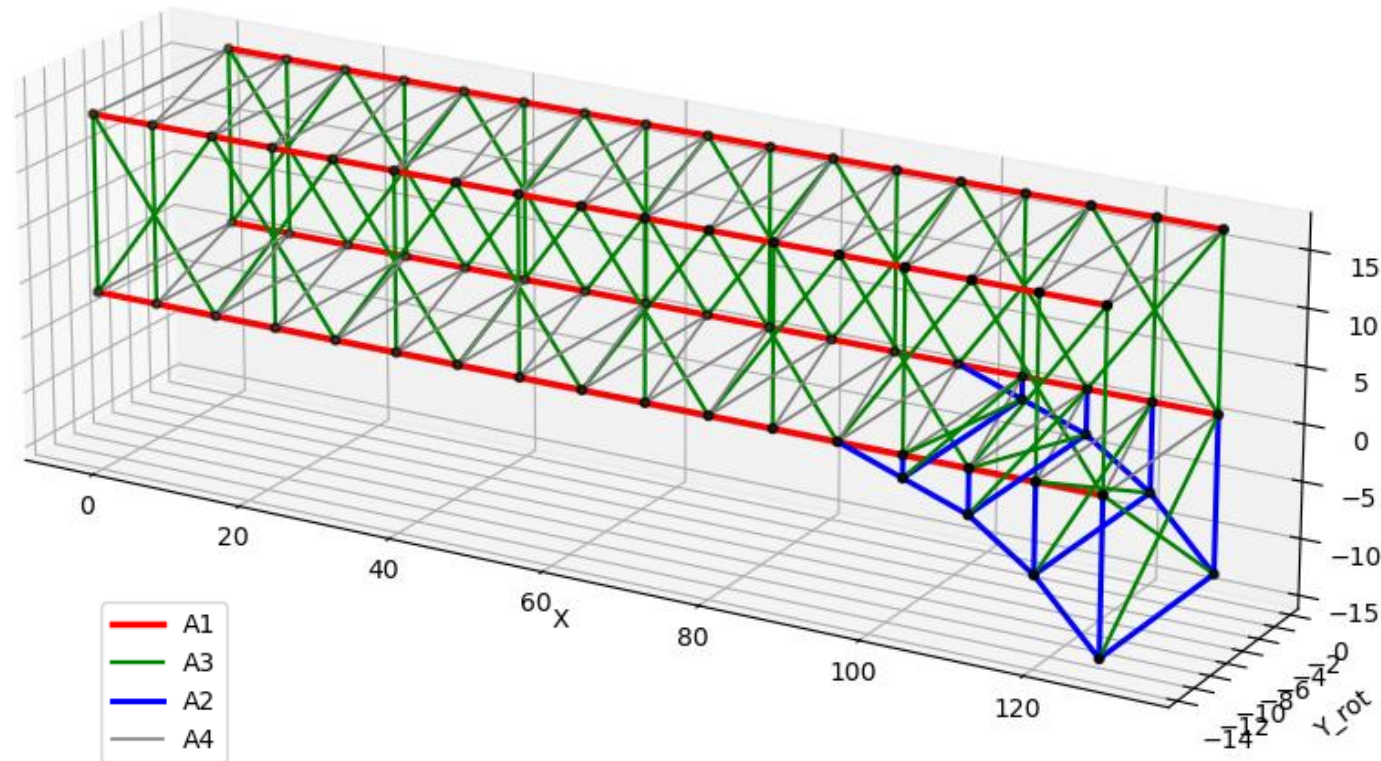
Description of the Final Structure

- Nanjing Yangtze River Bridge is selected as the target structure for the final part of the project.



Description of the Final Structure

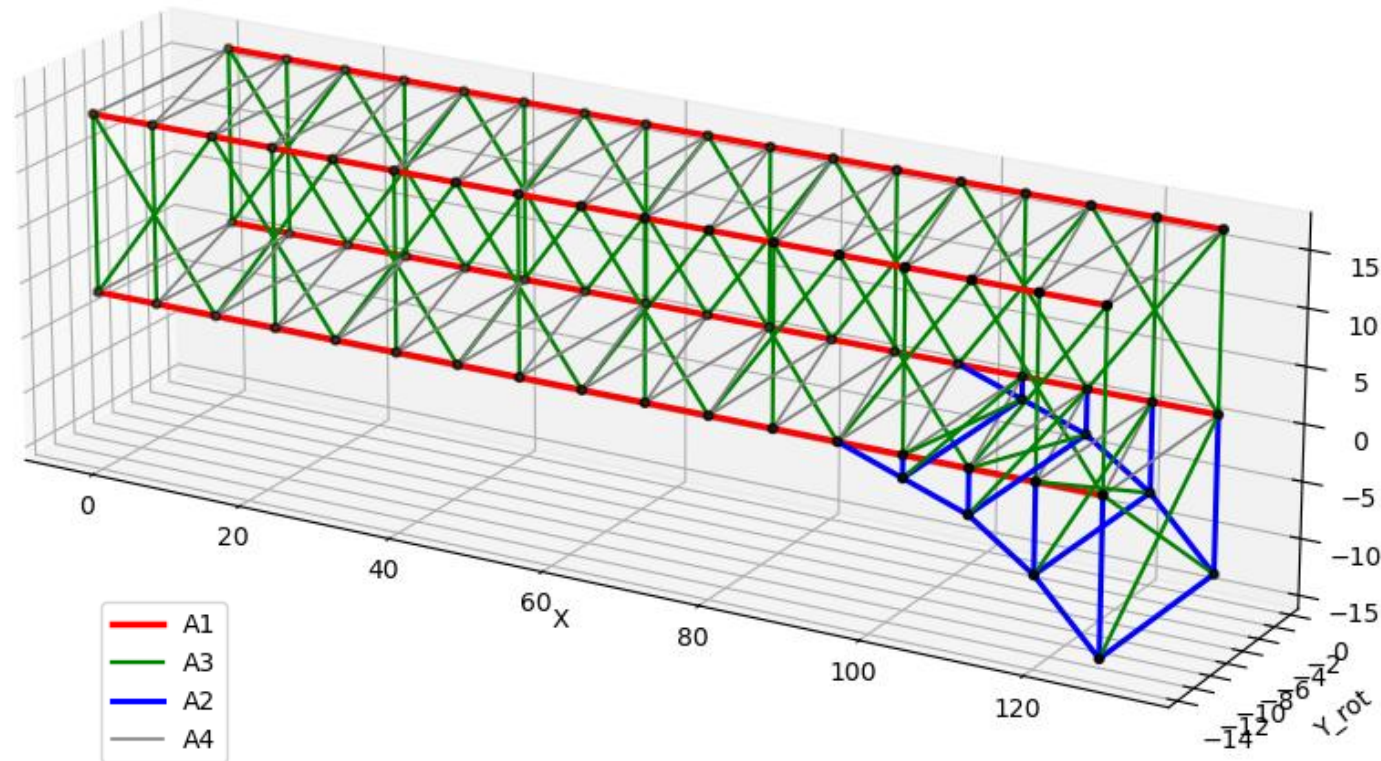
- Structural idealization: Truss members(A1, A3), Frame members(A2, A4)
- **Loading Assumptions**
- Vertical nodal loads → simulate self-weight + traffic
- UDL on lower deck beams (**member releases**) → represent railway loads
- Combined loading → captures global + local structural response



Description of the Final Structure

■ Advanced Effects Analysis

- Temperature ($\Delta T = 10^\circ\text{C}$) $\rightarrow$ global expansion
- Fabrication error ($\Delta L = -0.01\text{ m}$) $\rightarrow$ initial stress
- Support settlement ($u_y = -50\text{ mm}$) $\rightarrow$ local deformation



Analysis Results

▪ Global Behavior

Smooth and continuous **downward bending deformation** → No abnormal displacement → **physically consistent response**

▪ Displacement

- Max vertical: **-50.10 mm (Node 71)**

- Max longitudinal: **21.04 mm (Node 72)** → temperature effect → Vertical response dominates

• Rotation

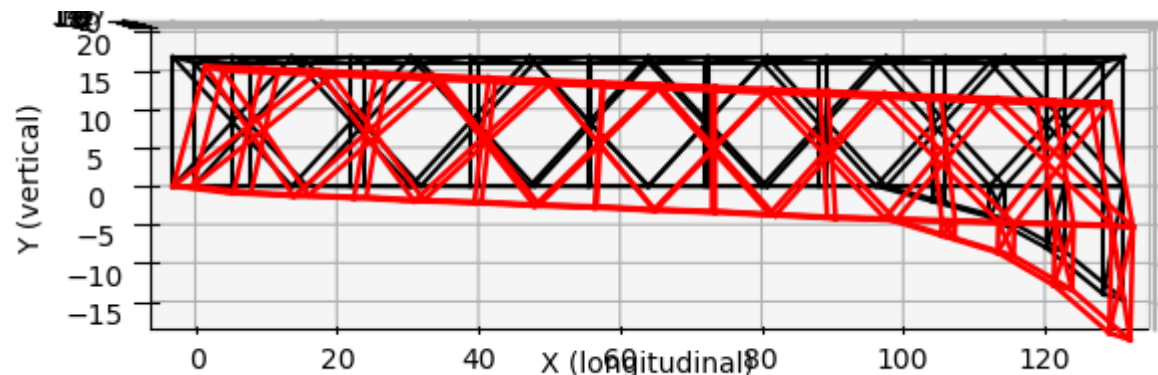
Very small ($\approx -4.46 \times 10^{-4}$ rad) → **High structural stiffness**, bending-dominated behavior

• Internal Forces

Axial-force dominated system (truss behavior) → Tension–compression distribution in members → Max stress $\approx$ **118 MPa (reasonable range)** → Consistent with DSM truss theory

• Support Reactions

Reactions match applied loads → **Global equilibrium satisfied** → Settlement causes redistribution (physically reasonable)



Conclusions

- Accurately predicts **displacements and internal forces**
- Results consistent with **engineering expectations**
- Captures effects of **temperature, fabrication error, and support settlement**
- Demonstrates **reliability and applicability** for bridge analysis
- Stable for **large-scale 3D mixed structures**

Thank You!

Questions/Comments?

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